Portfolio Management

BUS-F 303 INTERMEDIATE INVESTMENTS

Section 37012

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Portfolio Management Report

1. Portfolio Theory

1.1

I picked ten securities in different ranging from Information Technology to Auto Manufacturers. I based my decisions in trying to create a strategy where there are many market leaders and large cap companies in my portfolio such as Nike, Walmart, Starbuck, etc. I did add an ETF fund QQQ in my portfolio in order to further diversify my risks. After choosing my ten securities, I scraped the monthly price data (Adjusted close price) for each stock.

Info	ACN	SBUX	SONY	QQQ	BAC	MSFT	MA	GM	WMT	NKE
Industry	Information Technology	Restaurants	Consumer Electronics	ETF	Banking	Software	Credit Services	Auto Manufacturers	Discount Stores	Footwear
Founded	1989	1971	1946	2007	1904	1975	1966	1908	1962	1964
Market Cap	182.77B	133.989B	138.029B	153.76B	345.608B	1.93T	377.319B	85.965B	393.826B	215.887B
Price/Book Ratio	9.26	N/A	2.56	N/A	1.07	13.6	61.3	1.35	2.97	20
Beta	1.1	0.88	0.74	1.02	1.58	0.79	1.18	1.39	0.48	0.88

Figure 1.

1.2

From the monthly price date, I computed annualized monthly returns in order to calculate the average expected returns and standard deviation of each stock. The stock with the highest expected return was MasterCard and the stock with the highest standard deviation was Sony. Although the stock with the highest return does not also have the highest standard deviation, I can see a pattern that the more expected return a stock as the riskier it is.

Statistics	ACN	SBUX	SONY	QQQ	BAC	MSFT	MA	GM	WMT	NKE
Expected Return	20.45%	23.33%	16.68%	19.75%	14.63%	25.49%	30.38%	8.85%	13.45%	23.17%
Std. Dev.	68.20%	72.71%	118.93%	54.88%	113.24%	69.73%	72.28%	109.16%	59.73%	75.60%

Figure 2.

With ten stocks, I had to compute 45 pair-wise correlations. Using the Data Analysis tool, I was able to get a correlation matrix of all different combinations of my stocks. The pair of stock with the highest correlation was between Accenture and QQQ ETF. The pair of stock with the lowest correlation was between Walmart and Starbucks. When analyzing the information of each company's profile, we can expect to achieve good diversification benefit when the firm have completely different customer base, market strategy, and overall profile. For example, Walmart and Starbuck have great diversification benefit because Walmart prioritizes a low-cost strategy and Starbuck on the other hand relies on more of a high-quality strategy. Since these company's come from different industries, a market specific shock would only affect one stock greater than the other.

	ACN	SBUX	SONY	QQQ	BAC	MSFT	MA	GM	WMT	NKE
ACN	1.00									
SBUX	0.45	1.00								
SONY	0.35	0.23	1.00							
QQQ	0.73	0.45	0.50	1.00						
BAC	0.50	0.37	0.40	0.60	1.00					
MSFT	0.48	0.27	0.46	0.69	0.45	1.00				
MA	0.60	0.35	0.31	0.69	0.33	0.44	1.00			
GM	0.50	0.39	0.34	0.60	0.64	0.36	0.41	1.00		
WMT	0.27	0.08	0.12	0.23	0.08	0.13	0.10	0.17	1.00	
NKE	0.37	0.25	0.20	0.44	0.28	0.25	0.36	0.43	0.17	1.00

Figure 3.

1.3

I chose CAN, SBUX, SONY, QQQ, and BAC as the five stocks I wanted to create a portfolio from. I constructed my Mean-Variance Frontier by setting up a range of targeted portfolio returns.

I the solved the optimization problem of having the minimum standard deviation subject to the constraint of having the sum of weights equal to one and having the expected returns equal to the targeted portfolio returns. I did this optimization problem for each targeted portfolio returns in order to construct the Mean-Variance Frontier.
In order to get the global minimum variance portfolio, I used the same optimization problem as before but removed the constraint that the Expected Return had to equal the targeted portfolio return. When running solver, I got the global minimum portfolio to have an expected return of 21.28% and a standard deviation of 51%. Using the Sharpe Ratio formula for each portfolio, I plotted the highest one to the graph.

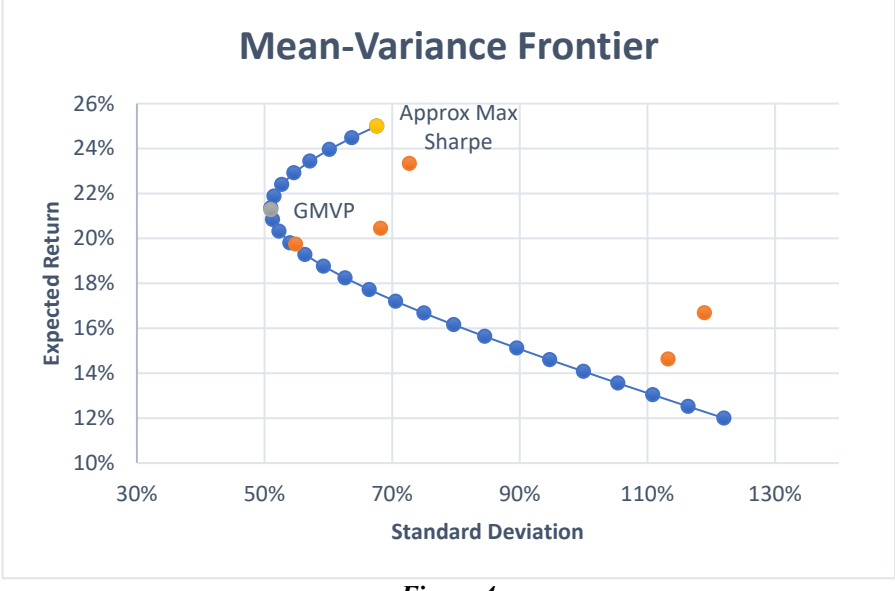

Figure 4.

2. CAPM Model

2.1

Doing the same arithmatic as I did for the stocks in my portfolio, I calculated the expected returns and standard deviation for the S&P 500 and Treasury Bond shown in ***Figure 5***. I then ran a regression for each of the stock in order to get the alpha, beta, and standard deviation of the residuals. Following that, I decomposed the expected return into the mispricing component and the required return component. I did the same for the risk, by decomposing it into the systematic risk and the idiosyncratic risk. MA had the highest total return, but BAC had the highest required return. MA also had the biggest mispricing historically. However, SONY had the highest total risk.

	S&P 500	Risk- Free
Expected Return	11.77%	0.58%
Std. Dev	47.07%	

Figure 5.

Stock	Alpha	Beta	Std. Dev of Residuals
ACN	7.61%	1.09	44.85%
SBUX	13.85%	0.80	62.60%
SONY	2.12%	1.25	103.78%
QQQ	7.25%	1.06	22.46%
BAC	-4.31%	1.64	83.17%
MSFT	15.19%	0.87	56.62%
MA	17.75%	1.08	51.70%
GM	-9.03%	1.54	81.75%
WMT	8.70%	0.37	57.29%
NKE	14.08%	0.76	66.91%

Figure 6.

Decomposed Returns			Decomposed Risk		
Expected Return	Required Component (CAPM Return)	Mispricing Component (Alpha)	Variance of Return	Systematic Risk	Idiosyncratic (Residual) Risk
20.45%	12.84%	7.61%	46.51%	26.56%	19.95%
23.33%	9.48%	13.85%	52.86%	14.03%	38.84%
16.68%	14.56%	2.12%	141.45%	34.56%	106.89%
19.75%	12.50%	7.25%	30.12%	25.11%	5.00%
14.63%	18.94%	-4.31%	128.24%	59.59%	68.65%
25.49%	10.31%	15.19%	48.62%	16.73%	31.89%
30.38%	12.63%	17.75%	52.25%	25.68%	26.57%
8.85%	17.87%	-9.03%	119.15%	52.87%	66.28%
13.45%	4.75%	8.70%	35.67%	3.09%	32.59%
23.17%	9.09%	14.08%	57.16%	12.80%	44.36%

Figure 7.

3. Portfolio Theory

3.1

In order to find the optimal risky portfolio, I did an optimization problem by setting the Sharpe Ratio as the objective we are trying to maximize and having the constraint where the sum of weights equals one. I then set up a arbitrary weight in the tangent portfolio against the Risk-free weights in order to set up a what-if analysis where we change the weight in the risk free. I then plotted all of the standard deviations and expected returns to get the Capital Market Line.

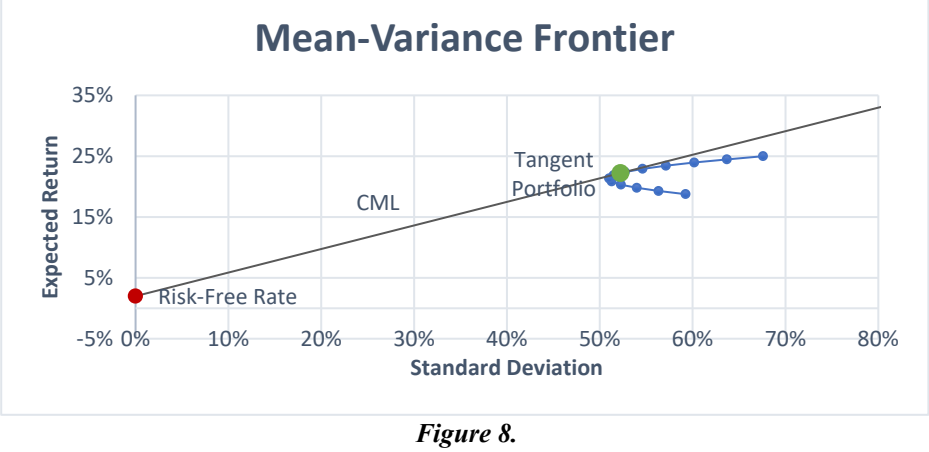

Figure 8.

3.2

In order to construct the mean-variance frontier where short selling of stocks is banned, I did the same process of creating an optimization problem. However, I added an additional constraint where each stock weight has to be greater or equal to zero. When combing the two frontiers, we see that there are a few portfolios that are aligned. I noticed that they have contact mostly near the global minimum variance portfolio and then start to diverge. The impact a short selling constraint have on portfolio diversification is that the constrain does not allow the mean-variance frontier to maximize the returns to risk. When we allow short selling, we allow correlations to have their full effect to produce the most efficient frontier.

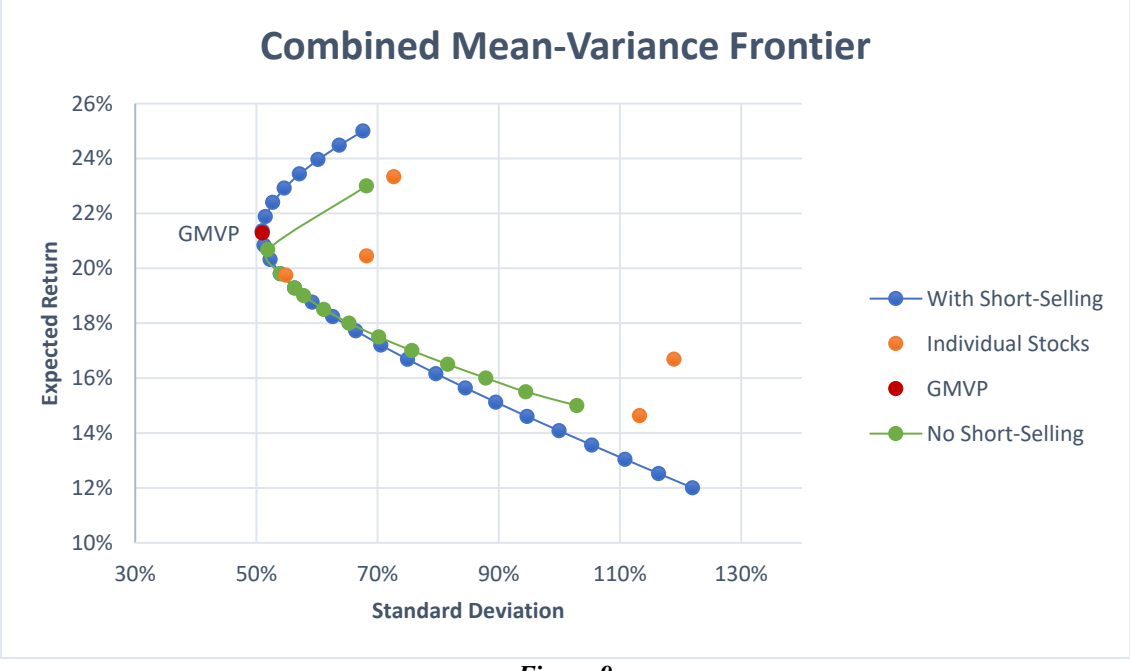

Figure 9.